

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Tutorial Sheet 6

23 and 24 March 2023

1. Let the vector function $\vec{r}: [-\sqrt{5}, 3] \rightarrow \mathbb{R}^2$ be differentiable at -1 with $\vec{r}(-1) = (-1, \pi)$ and $\vec{r}'(-1) = (-\sqrt{2}, \frac{1}{3})$. If the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable at the point $(-1, \pi)$ with $f_x(-1, \pi) = 2\sqrt{2}$ and $f_y(-1, \pi) = 6$, then explain why the function $f \circ \vec{r}$ is differentiable at -1 find the derivative $(f \circ \vec{r})'(-1)$.
2. (i) Find the directional derivative of $f(x, y, z) = \sin x \cos y \tan z$ at the point $(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3})$ in the direction $\vec{u} = (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})$.
(ii) Find the directional derivative of $f(x, y, z) = 2x^2 - y^2 + z^3 - 2xyz + x + 3y + 1$ at the point $(1, 1, 1)$ in the direction of the unit tangent vector of the curve $\vec{r}(t) = (\cos t, -\sin t, t)$ at the point where $t = -\frac{\pi}{4}$.
3. (i) Given the function $f(x, y) = 2x^2 + y + 3y^2 - 1$ and the unit vector $\vec{u} = (\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$. Find all points (x, y) such that the directional derivative $D_{\vec{u}}f(x, y)$ is equal to 1.
(ii) If g is a function differentiable at (a, b) such $\nabla g(a, b) = (2, 3)$, then find all the vectors $\vec{u} = (x, y)$ with $x^2 + y^2 = 1$ such that $D_{\vec{u}}g(a, b) = 0$.
4. Let $\vec{u} = \frac{1}{\sqrt{5}}\langle 2, 1 \rangle$ and $v = \frac{1}{\sqrt{2}}\langle 1, -1 \rangle$. If the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is differentiable at the point (a, b) with $f_{\vec{u}}(a, b) = -2$ and $f_{\vec{v}}(a, b) = 1$, then find:
(i) $\nabla f(a, b)$;
(ii) the maximum increase of the function f at the point (a, b) .